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Orbital Satellite Communication Encryption Algorithm Based on Elliptic Curve Cryptography Algorithm

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Abstract

In orbital satellite communication systems, there is a large amount of user sensitive data. In order to ensure the security of the data, but also to ensure the data transmission rate, the general use of symmetric encryption algorithm. On this basis, a new encryption method based on elliptic curve is proposed. It is a new encryption method based on elliptic curve. It is a new encryption method based on elliptic curve. Simulation results show that the encryption method proposed in this paper is feasible, and its security is better than 98%. It can be applied to in-orbit satellite communication.

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Keywords: communication encryption, elliptic curve cryptography, orbiting satellite, RSA algorithm

1. Introduction

At present, with the continuous development of satellite communication technology, higher requirements for communication data security are put forward. Traditional encryption methods to ensure information security, there are also some shortcomings, such as low efficiency, low security and so on. In this paper, the elliptic curve cryptography is studied deeply, and an encryption method with great application prospect is proposed.

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Firstly, the theory of elliptic curve cryptography is expatiated in detail, and the application of elliptic curve cryptography in satellite communication is studied. And its security is analyzed. Finally, the method is compared and analyzed, and the results show that the method is safe.

2. Related work

In recent years, there have been many studies on communication encryption. For example, Visconti P studied a high throughput data processing and encryption technology solution for 5G communication [1]. Arthi G proposed a combination of 4D hyper chaotic mapping and DNA encoding for image encryption to achieve secure communication [2]. Rajasekar A applied raw data encryption technology to communication networks [3]. Dankan Gowda V constructed a secure communication model for intelligent binary image encryption using vector space modeling [4]. Zehtabchi S proposed a new method for privacy preserving association rule mining based on homomorphic encryption and secure communication protocols [5]. The security of encryption algorithms in these studies is very good, but empirical research is insufficient.

There are also many studies on the application of ECC algorithms. Khan M R analyzed the characteristics of ECC [6]. Kumar A applied ECC to smart grid communication systems [7]. Velliangiri S proposed an efficient and lightweight privacy protection mechanism for Industry 4.0 based on ECC [8]. Perumal P studied a secure communication analysis of wearable healthcare systems based on ECC [9]. Oladipupo E T improved the authenticated ECC scheme for resource scarce applications [10]. The above content provides a thorough analysis of the application of ECC algorithms, but it has not been comprehensively compared and analyzed with other algorithms.

3. Methods

3.1. ECC Algorithm

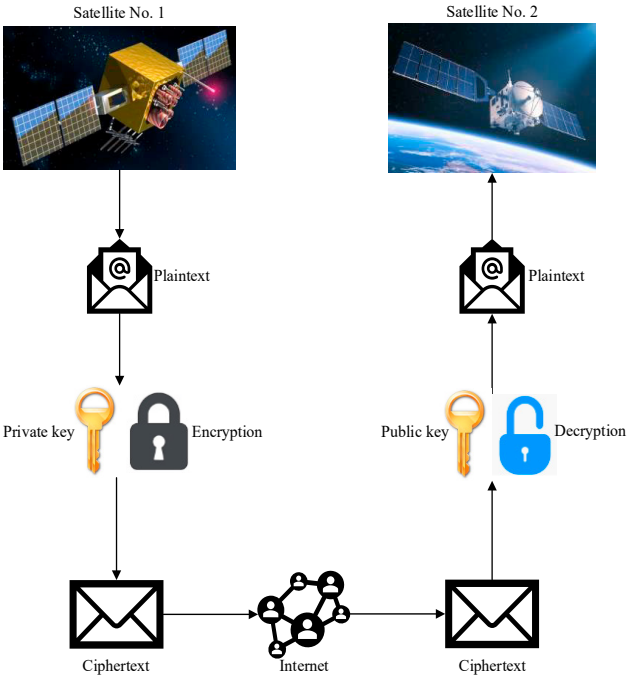


Fig 1. Running process of ECC algorithm

The ECC algorithm is applied in fields such as identity authentication, electronic signatures, and digital signatures. Its biggest advantages are fast algorithm speed, high encryption strength, strong anti attack ability, and easy implementation. The elliptic curve cryptosystem has extremely high computational efficiency and security, so it is also known as the fastest public key encryption algorithm. The ECC algorithm consists of a public key algorithm and a private key algorithm, as shown in Figure 1.

In Figure 1, the data of satellite 1 is first encrypted using ECC; then, the encrypted data is transmitted over the network and decrypted using the ECC algorithm; finally, the information is transmitted to the second satellite. In ECC algorithms, points on the elliptic curve are used as private and public keys. This public key system can be widely used in fields such as identity authentication, electronic signatures, and digital signatures [11].

The ECC algorithm is a symmetric encryption system based on elliptic curves, and its security depends on the number of points on the elliptic curve, the relative position relationship between points, and the point multiplication operation on the elliptic curve. In the ECC algorithm, the private key a is generated by the user. The user stores the private key a and the public key b in two different files. The private key a is encrypted by an ECC algorithm to form a set of asymmetric encryption for encrypting sensitive user data; the public key b is encrypted by an ECC algorithm to form a set of asymmetric encryption for secondary encryption of user sensitive data.

The user stores their public key b in a file, which can be a pure article file or a pure digital signature file. Among them, pure articles contain all communication information between users and the outside world; the digital signature file contains the signature information of all communication information between the user and the outside world. ECC algorithm is a secure and efficient symmetric cryptographic algorithm. In general, its computational efficiency is much higher than that of public key cryptographic algorithms, and it can effectively resist various attack methods. However, in practical applications, due to the enormous computational complexity of elliptic curves and the complexity of calculating points on elliptic curves, elliptic curve algorithms are generally not suitable for processing applications with high computational complexity and slow computational speed. In addition, the ECC algorithm has strict requirements for key length, which can greatly affect its efficiency when the key length is relatively short. Therefore, it is necessary to improve it to enhance its security [12].

The ECC in ECC algorithms is used to protect user private keys. ECC is a novel key generation algorithm that represents keys of any length and known length in the form of discrete logarithms, and uses modular operations to transform these discrete logarithms into random integers of length 1. A public key cryptosystem based on elliptic curve cryptosystem is proposed. In the elliptic curve encryption algorithm, the length of the key is reduced through modular computation; In the elliptic curve encryption system, the public key can be generated without repeated computation for keys of the same length; and The public key generated by the elliptic curve encryption system can be of any length and length. Therefore, the elliptic curve encryption algorithm can effectively enhance the network security. However, the ECC algorithm also has its drawbacks: (a) it requires a large amount of computing resources during key generation; (b) the ECC algorithm has a slower computational speed.

3.2. Security

In the article, the encryption algorithm of elliptic curves can be used to encrypt user sensitive data in orbital satellite communication systems. The basic idea of ECC is to encrypt data by solving elliptic curve equations, calculate a key using a one-way hash function, and then use this key for decryption. The elliptic curve cryptosystem has the following advantages: the calculation of any point on the elliptic curve can be obtained through encryption and decryption algorithms. Due to the fact that the private key and public key are two different points on the same piece of paper, even if an attacker has a large number of public keys, they cannot decipher the information contained in the private key [13].

Due to the fact that the dot product corresponding to any point on an elliptic curve is a dot product of any length, ECC has strong computational power. The longer the key length, the less sensitive it is to plaintext and ciphertext, because the ciphertext length used in public key encryption systems is fixed. As long as the length of the private key used in the public key cryptography is known, the relationship between plaintext and ciphertext under the public key cryptography can be obtained. The ECC algorithm does not need to know any relevant information between plaintext and ciphertext, so it can greatly increase the key length. Key update mainly updates the parameters at each point on the elliptic curve. Due to the fact that ECC algorithms require a significant amount of time to find and

calculate private and public keys, there is a significant time difference between plaintext and ciphertext. This results in the time required to generate keys in ECC algorithms being much smaller than in public key cryptography.

In the implementation process of ECC algorithm, a private key must be passed between the receiver and sender. When the sender sends encrypted data to the receiver, the receiver must send a private key to the sender, and the receiver uses this private key to encrypt and decrypt the plaintext to ensure that the data is not stolen or tampered with during transmission. The public and private keys in ECC are two points on the same piece of paper, making key management very difficult. Due to the absence of a public key algorithm in ECC, it is difficult to obtain specific information about the private and public keys used by all users in the system.

Message integrity refers to the fact that information is not tampered with or forged during transmission. Due to the fact that the communication object of the orbital satellite communication system is a massive amount of data, in order to ensure the security of user information, it is necessary to encrypt it. Firstly, a hash function is used to encrypt the message, as in traditional information transmission, the hash value of the message is set and saved locally by the receiver and sender. The receiver can only ensure that their message is not tampered with by obtaining the hash value. Secondly, in the traditional information transmission process, for the sender, they can only know what content they have sent, while for the receiver, they can know what content they have received. Therefore, if the recipient needs to encrypt the message, they would not be able to know the specific content of the message. However, when using ECC algorithm to encrypt messages, the receiver can know what data they have received and how this data has been transmitted to communication satellites. Therefore, the integrity of the message can be determined by the hash value of the receiver. If the recipient is using ECC algorithm for encryption, then they use the recipient's own private key for decryption. It is assumed that the recipient decrypts the message using their own private key and obtains a message digest. If the recipient encrypts the message digest with their own public key and obtains a plaintext message, the decryption equation for the public key specified in the ECC algorithm is [14]:

$$\begin{aligned} M &= x_1 - p_u y_1 \\ M &= x_1 - p_r y_1 \end{aligned} \tag{1}$$

Among them, M is plaintext; x_1 is ciphertext; p_u is the public key; p_r is the private key; y_1 is the basis point. When a user A needs to encrypt their data, first, user A needs to apply to the network operator for a public key and a private key. Then, user A encrypts the data using the public key and decrypts it using the private key. During this process, if user A wants to modify their data, they need to use a private key to decrypt the data. To ensure that data is not tampered with or stolen during transmission, network operators verify the public key of user A, and only the verified public key can be used for encryption.

Finally, user B verifies the decrypted data using a private key. When user B encrypts their data using a public key, the verification process is similar to the above. If there is an issue during this process, it indicates that user B's public key does not come from the public key requested by the network operator. Meanwhile, if the public key applied for by the network operator is inconsistent with the public key used by user B, it indicates that user B has a public key that is not from the network operator's application.

The confidentiality of public key cryptography mainly lies in the confidentiality of keys and messages. The confidentiality of keys mainly refers to the generation of public keys through private keys, which cannot be directly used for encryption; the confidentiality of messages refers to the fact that only private keys can decrypt and confirm messages, while public key algorithms can ensure that messages are not eavesdropped during transmission.

The security of ECC algorithm depends on whether it can be encrypted and decrypted securely, as it adopts two steps: encryption and decryption. To ensure the security of ECC algorithms, it is necessary to study the following aspects: computational efficiency is one of the most important factors affecting the security performance of the entire cryptographic system, and it is also an important indicator for measuring the security performance of a cryptographic system. Low computational efficiency means that the ECC algorithm has high computational complexity, which results in long and complex security analysis of the algorithm, affecting its practical application. Therefore, in practical applications, it is necessary to improve the computational efficiency of the ECC algorithm as much as possible.

In ECC, communication authentication is only required when transmitting data between the receiver and sender. This is because when transmitting data between the receiver and the sender, the sender encrypts the public and private keys before sending them to the receiver, and the receiver needs to decrypt the information to confirm its correctness. Therefore, communication authentication is only required between the sender and receiver when transmitting data in both the receiver and the sending direction. Due to the use of the same key for encryption and decryption in ECC algorithms, even if the private keys of the receiver and sender are the same, as long as the communication parties use different public keys to encrypt and decrypt information, the correctness of the message cannot be guaranteed. Due to the fact that ECC algorithms require information to be transmitted through channels during data transmission, channel integrity has become an important factor in ensuring the security of ECC algorithms [15].

4. Experimental Results and Discussion

At the end of the article, a series of experiments were conducted, first focusing on the core content of encryption algorithms - security. The ECC algorithm used in the article was compared and analyzed with the relatively advanced RSA (Rivest-Shamir-Adleman) algorithm. The results are shown in Figure 2.

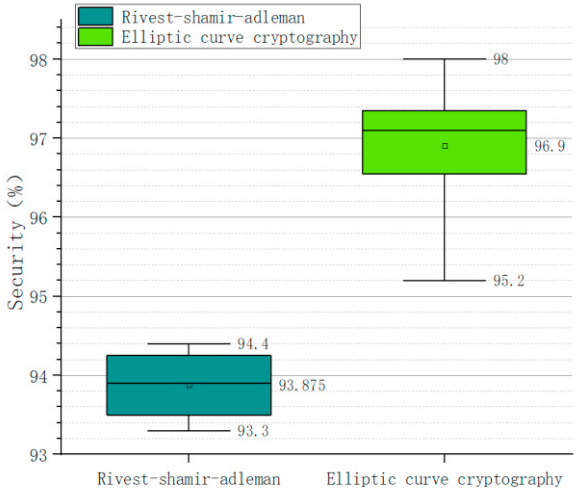


Fig 2. Security

In Figure 2, the security of the RSA algorithm reached a maximum of 94.4% and a minimum of 93.3%, and the calculated average security was 93.875%; the security of ECC algorithm was highest at 98% and lowest at 95.2%, with an average calculated security of 96.9%. It can be seen that the ECC algorithm used in the article had higher security performance and could better safeguard satellite data transmission. Subsequently, the article conducted a robustness analysis, as shown in Figure 3.

In Figure 3, the highest robustness of the RSA algorithm was 88%, which appeared in the fourth experiment, and the lowest was 85.1% in the eighth experiment. The calculated average robustness was 86.6%; the maximum robustness of the ECC algorithm reached 92.8%, which appeared in the 8th experiment, and the lowest was 90.3% in the 4th experiment. The calculated average robustness was 91.55%. Subsequently, the article collected data from other encryption algorithms and analyzed their probability of being attacked. The results are shown in Table 1.

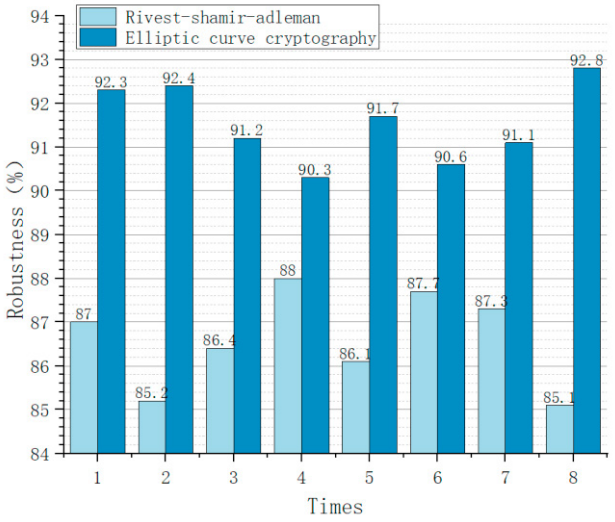


Fig 3. Robustness

Table 1. Probability of being attacked

Encryption algorithm	Attack method		
	Interfere	Tamper	Steal
Advanced encryption standard (AES)	0.23	0.25	0.15
Rivest-shamir-adleman (RSA)	0.18	0.16	0.17
ECC (ECC)	0.09	0.08	0.05
Secure hash algorithm (SHA)	0.15	0.17	0.13
Keyed-hashing for message authentication (HMAC)	0.21	0.19	0.17

In Table 1, the AES algorithm (0.23) had the highest probability of interference; the AES algorithm (0.25) had the highest probability of being tampered with; the RSA algorithm and HMAC algorithm (0.17) had the highest probability of being stolen. The ECC algorithm had the lowest probability in any attack mode. Finally, the article also studied the compatibility of satellites. The results are shown in Table 2.

Table 2. Compatibility

Classification of orbital satellites	Compatibility classification		
	Spectrum	Signal interface	Ground equipment
Geostationary earth orbit	85.8%	87.0%	87.4%
Low earth orbit	81.4%	84.6%	85.1%
Medium earth orbit	88.0%	86.0%	82.8%
Highly elliptical orbit	86.7%	84.8%	88.6%

In Table 2, the spectrum compatibility of medium orbit satellites was the highest at 88%; the signal interface compatibility of geostationary orbit satellites was the highest at 87%, and the ground equipment compatibility of highly inclined orbit satellites was the highest at 88.6%. Of course, these data are not absolute, they are only test data from the satellites selected in the article.

The article studied the application of ECC algorithm in satellite communication in orbit. The ECC algorithm is an efficient and secure encryption method suitable for use in satellite communications with limited space. Applying ECC technology to orbital satellite communication can ensure the confidentiality and integrity of information. The article provided a detailed analysis of the ECC algorithm used in orbit satellite communication, studied a set of encryption algorithms suitable for in orbit satellite communication, and evaluated and analyzed their security and robustness to verify their application value in orbit satellite communication.

5. Conclusions

The article studied the application of ECC algorithm in orbital satellite communication systems, using elliptic curve as the private key and symmetric cryptography as the public key to encrypt sensitive user data, and achieving secure information transmission through secondary encryption. Compared with other existing encryption algorithms, this algorithm had higher security and could achieve higher computational speed. Due to the efficiency of discrete logarithm operations on elliptic curves, elliptic curve algorithms can be used instead of symmetric cryptographic algorithms in orbital satellite communication systems. However, due to the use of public keys for encryption in elliptic curve algorithms, there are certain requirements for the efficiency of discrete logarithm operations on elliptic curves.

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